

In one scenario, the computer predicted that the U.S. stock market would fall by 15% in one month. The computer certainly didn't give a reason. It didn't explain why that would happen. It simply said that it was within the realm of possibility. And lo and behold, during August of 1998, the S&P 500 index fell by roughly that amount. Quite uncanny!

Obviously, this doesn't mean that the computer actually predicted the future. It simply means that, using the Monte Carlo simulation, it computed a remote possibility of the market's declining by 15% in one month—and, implicitly, recommended that we should plan for such a possibility. I don't want to alarm you unnecessarily, but in a few scenarios it generated drops even scarier than 25%. Fortunately, the computer estimated the odds of such events to be very small.

Now comes the fun part. After leaving the computer on all night, running millions of these future scenarios, we returned in the morning and started counting.

Specifically, for every possible asset allocation, we counted the number of times that the 65-year-old who starts out with \$200,000 and consumes \$16,000 per year will run out of money before she dies. These are the people who (theoretically) run out of money. The remaining people, who die with wealth, have managed to avoid outliving their money. The ratio of the former to total number of trials provides us with the probability of outliving your money. We then sift through the cases to locate the asset allocation that minimizes the probability of outliving wealth.

Let's look at an example. In one simulation, a 65-year-old who invested all of his capital in the stock market—consuming \$16,000 real dollars each year—ran out of money in roughly five years because he had the uncanny bad luck of investing right before a horrendous (computer-generated) bear market. Again, however, the computer assigned a very low probability to this event.

In another scenario, we found that a 55-year-old was able to take very early retirement, with only \$200,000 in wealth, and still manage to consume \$16,000 per year for life, by investing completely in equities. But this is because the computer killed him off at age 68, long before his full life expectancy. The computer also assigned this event—a